

Advanced (Habanero)

Q1. Solve for x in the equation $2^x = 8$

- (A) $x = 2$
- (B) $x = 3$
- (C) $x = 4$
- (D) $x = 5$
- (E) $x = 6$

Q2. Solve for y in the equation $3^y = 27$

- (A) $y = 1$
- (B) $y = 2$
- (C) $y = 3$
- (D) $y = 4$
- (E) $y = 5$

Q3. Solve for z in the equation $4^z = 64$

- (A) $z = 1$
- (B) $z = 2$
- (C) $z = 3$
- (D) $z = 4$
- (E) $z = 5$

Q4. Solve for x in the equation $2^{x-1} = 4$

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Q5. Solve for y in the equation $5^{y+1} = 125$

- (A) $y = 1$
- (B) $y = 2$
- (C) $y = 3$
- (D) $y = 4$
- (E) $y = 5$

Q6. Solve for z in the equation $2^{z+2} = 32$

- (A) $z = 1$
- (B) $z = 2$
- (C) $z = 3$
- (D) $z = 4$
- (E) $z = 5$

Q7. Solve for x in the equation $3^{x-2} = 9$

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Q8. Solve for y in the equation $4^{y-1} = 16$

- (A) $y = 1$
- (B) $y = 2$
- (C) $y = 3$
- (D) $y = 4$
- (E) $y = 5$

Open Ended Questions (FRQ)

Q9. Solve for x in the equation $2^x + 3^x = 90$

Q10. Solve for y in the equation $4^y - 2^y = 60$

Chef's Tips

- Check for correct use of exponent rules
- Ensure students show all work and explain their reasoning
- Be available to assist students as needed